Surrogate models for Monte Carlo uncertainty propagation with few runs of an expensive model: specification, selection and a valid surrogate contribution

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Abstract

When the measurement model is an expensive simulation, Monte Carlo propagation of distributions according to GUM Supplement 1 is carried out on a surrogate fitted to a few tens of model runs, and the surrogate's error becomes part of the uncertainty budget. We compare Gaussian processes (GPs) of different specifications, sparse polynomial chaos expansions, committees of surrogates, and space-filling, adaptive and threshold-targeted designs on ten models, including GUM-type input distributions and a computational fluid dynamics (CFD) simulation of a gas release behind a building, with 10 to 80 runs and 30 paired repetitions against reference Monte Carlo evaluations. The specification of the surrogate mattered more than the choice of strategy: an estimated kernel amplitude, a quadratic trend and the log scale for positive quantities reduced the error by factors of up to 18, 400 and 9, and choosing among such candidates by their leave-one-out error recovered practically all of the gain without access to the true model. Committees did not beat a GP with a trend, and variance-driven adaptive designs gave no consistent gain at 20 runs and helped only with larger budgets on smooth models, whereas a design targeted at a threshold in the tail reduced the error of exceedance probabilities up to ninefold. Adding the leave-one-out residuals of the surrogate to every Monte Carlo trial brought the achieved coverage of the GUM S1 95% coverage interval to 91–97% on all models, and jackknife+ prediction intervals covered 95–99% of test points, whereas the GP's own predictive intervals covered as few as 68%, and its intervals for a 95th percentile contained the true value in a median of only 53% of repetitions. The results are condensed into a validated procedure for surrogate-based uncertainty evaluation.

Keywords: measurement uncertainty; GUM Supplement 1; Monte Carlo method; surrogate model; Gaussian process; polynomial chaos; conformal prediction; gas dispersion.

1 Introduction

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1, 2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial, typically 10^5 to 10^6 times [3, 4]. For a growing class of quantities f is a numerical simulation. A computational fluid dynamics (CFD) model of an accidental ammonia release in a factory [5] can take hours to days per run, so that a practical budget is a few tens of runs, three or four orders of magnitude short of what the Monte Carlo method needs. The same gap arises in scatterometry, where it has been studied in detail [6], and in many inverse and model-based measurements.

The usual remedy is a *surrogate* (emulator, metamodel) fitted to the available runs and evaluated in place of f [7, 8]. Two questions then decide whether the resulting uncertainty statement can be trusted: which surrogate reproduces the reported quantities best for a given number of runs, and how the surrogate's own error is to be included in the uncertainty budget, since it is not negligible at such budgets. This paper answers both with a controlled comparison on ten models, including a CFD dispersion case, and validates the answers against the true models.

1.1 Related work

Gaussian process emulators. Gaussian process (GP) regression, or kriging, is the standard surrogate for deterministic computer experiments [9, 10]. The Bayesian treatment of O'Hagan and co-workers [7, 11] reads the GP posterior as the uncertainty about f itself, which suggests using the posterior variance as the surrogate contribution to an uncertainty budget. A common guideline asks for about ten runs per input [12], and the effect of modelling choices such as the correlation family, the mean function and the treatment of the hyperparameters on GP accuracy has been assessed systematically [13].

Polynomial chaos. Polynomial chaos expansions (PCE) represent Y in polynomials orthonormal with respect to $g_{\mathbf{X}}$ [14], from which the moments and sensitivity indices follow directly [15]. With few runs, sparse expansions selected by least-angle regression [16] and a corrected leave-one-out error are the method of choice [17].

Ensembles and committees. Combining several surrogates with weights derived from their cross-validation errors protects against choosing a poor surrogate [18, 19]; the spread of the members has also been used as an interval for the output, in the spirit of probability boxes [20].

Adaptive designs. Instead of fixing all runs in advance, runs can be added where the surrogate is most uncertain, by its posterior variance [21], by the disagreement of a committee, or by measures such as the spectral uncertainty of Lin et al. [22]. For the probability of exceeding a threshold, the AK-MCS method of Echard et al. [23] places runs where the sign of $f - T$ is most uncertain.

Validity of the surrogate contribution. Whether any of these uncertainty descriptions has the coverage it claims is rarely checked against the true model. Conformal prediction [24], and in particular the jackknife+ [25], gives prediction intervals with a coverage guarantee that holds for any fitted model, using only leave-one-out fits. To our knowledge it has not been compared with GP posterior intervals as the surrogate contribution in a GUM S1 evaluation.

Benchmarks. Comparisons of surrogates are sensitive to the test functions and to the number of repetitions. We use standard functions from the literature [26, 27], GUM-type input distributions, a physical dispersion formula and a CFD case, with 30 paired repetitions and statistics adjusted for multiple comparisons.

1.2 Earlier analysis and contributions

An earlier analysis by the first author combined a heterogeneous committee, the spectral acquisition criterion and dynamic model management, and reported that a passively sampled GP fails catastrophically on large input domains, a failure the committee avoided. The present study re-examines these claims with correctly specified baselines and extends them to the general question above. Its contributions are:

1. a paired comparison, on ten models with uniform and GUM-type inputs and a CFD dispersion case, of GP specifications, sparse PCE, committees and adaptive and targeted designs at budgets of 10 to 80 runs, scored on the quantities a GUM S1 evaluation reports;

2. the finding that the GP failure of the earlier analysis is a kernel-specification artefact, and that specification (amplitude, anisotropy, trend) and variable transformation matter more than the choice between single surrogates, committees and adaptive designs;
3. a validation, against the true models, of three ways to include the surrogate error in GUM S1: the GP posterior, a cross-validation (CV) normal error and resampled leave-one-out residuals, with jackknife+ intervals as a distribution-free reference;
4. a procedure for surrogate-based GUM S1 evaluations (Fig. 6), and code, per-seed records and an audit script that recompute every number reported [28].

2 Surrogate-based propagation and the surrogate contribution

2.1 Propagation of distributions through a surrogate

Let $Y = f(\mathbf{X})$ be the measurement model, $g_{\mathbf{X}}$ the joint probability density of the inputs (here with independent components) and $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, $y_i = f(\mathbf{x}_i)$, the N available model runs. A surrogate $\hat{f}$ is fitted to the runs and the Monte Carlo method of GUM S1 [2] is applied to $\hat{f}$: M draws $\mathbf{x}^{(r)} \sim g_{\mathbf{X}}$ give $\hat{y}^{(r)} = \hat{f}(\mathbf{x}^{(r)})$, from which the estimate y (the mean), the standard uncertainty $u(y)$ (the standard deviation), the probabilistically symmetric 95% coverage interval $[y_{0.025}, y_{0.975}]$, the 95th percentile $y_{0.95}$ and the probability $P(Y > T)$ of exceeding a threshold T are computed. We call these the *reported quantities*.

Surrogates are fitted on a *latent* scale $z = \varphi(y)$, with $\varphi = \log$ for the log-variable variants and the identity otherwise, and predictions are mapped back by φ^{-1}. For a Gaussian latent predictive distribution, exp of the latent mean is the predictive median of Y; no bias correction is applied.

2.2 The surrogate error as an uncertainty contribution

The surrogate error $e(\mathbf{x}) = \varphi(f(\mathbf{x})) - \varphi(\hat{f}(\mathbf{x}))$ is unknown. Following Kennedy and O'Hagan [11], who treat model inadequacy as an uncertainty source, we propagate

$$Y = \varphi^{-1}\Big(\varphi(\hat{f}(\mathbf{X})) + E\Big), \tag{1}$$

with E a random variable describing the surrogate error, drawn independently for every Monte Carlo trial. Three models for E are compared, all available from the N runs alone:

- *GP*: $E \mid \mathbf{x} \sim \mathrm{N}(0, s^2(\mathbf{x}))$, with $s(\mathbf{x})$ the predictive SD of a Gaussian process surrogate;
- *CV-normal*: $E \sim \mathrm{N}(0, s^2_{\mathrm{LOO}})$, with $s^2_{\mathrm{LOO}} = N^{-1} \sum_i r_i^2$ and $r_i = \varphi(y_i) - \varphi(\hat{f}_{-i}(\mathbf{x}_i))$ the leave-one-out (LOO) residuals; $\hat{f}_{-i}$ is the surrogate re-fitted, hyperparameters included, without run i;
- *resampling*: E drawn with equal probabilities from $\{r_1, \ldots, r_N\}$.

The same ingredients give pointwise 95% prediction intervals for $f(\mathbf{x})$: the GP interval $\hat{z} \pm 1.96\, s(\mathbf{x})$, the CV-normal interval $\hat{z} \pm 1.96\, s_{\mathrm{LOO}}$, and the jackknife+ interval [25]

$$\left[\, q^-_\alpha\{\hat{z}_{-i}(\mathbf{x}) - |r_i|\},\ q^+_\alpha\{\hat{z}_{-i}(\mathbf{x}) + |r_i|\} \,\right], \tag{2}$$

where q^-_α is the $\lfloor \alpha(N+1) \rfloor$-th smallest and q^+_α the $\lceil (1-\alpha)(N+1) \rceil$-th smallest of the N values, $\alpha = 0.05$. For exchangeable runs the jackknife+ interval covers f with probability at least $1 - 2\alpha$ whatever the surrogate [25]; it is a member of the conformal family [24].

Table 1: Test models. d: number of inputs; U: independent uniform inputs.

Model	d	Inputs	Character
Physics	3	U$[-200, 200]^3$, noise SD 0.1	quadratic bowl plus small oscillatory terms
Plume	5	U: Q, u, H, x, y	multiplicative, positive, heavy right tail
Borehole	8	U, ranges of [26]	smooth, multiplicative, positive
Ishigami	3	U$[-\pi, \pi]^3$	strongly non-linear, interactions
OTL circuit	6	U [31]	smooth, nearly additive, positive
Piston	7	U [31]	non-linear, positive
Borehole (GUM)	8	normal, lognormal, uniform [29]	as borehole; input tails
Plume (GUM)	5	lognormal, truncated Weibull, uniform, normal	as plume; heavier input and output tails
Flood	8	truncated Gumbel and normal, triangular, uniform [32]	smooth, lower-tail threshold
CFD dispersion	6	U: U_{ref}, z_0, H_{b}, Sc_t, x_r, Q	RANS solution: wake recirculation, numerical error

For a reported quantity θ itself (the mean, SD, $y_{0.95}$ or $P(Y > T)$ of the output over a fixed set of input draws), two 95% intervals are compared: the 2.5th and 97.5th percentiles of θ over joint sample paths of the GP posterior, and the jackknife interval $\hat{\theta} \pm 1.96\,\mathrm{SE_J}$ with $\mathrm{SE_J^2} = \frac{N-1}{N}\sum_i(\hat{\theta}_{-i} - \bar{\theta})^2$, where $\hat{\theta}_{-i}$ is computed from $\hat{f}_{-i}$.

3 Test models

Table 1 lists the ten models. Six have independent uniform inputs, three have the non-uniform input distributions of a GUM-type evaluation, and one is a CFD simulation. The physics model is the benchmark of the earlier analysis; its quadratic term carries practically all the output variance on $[-200, 200]^3$ and lies in the class of a degree-2 polynomial. The plume model is the ground-level concentration of a continuous ammonia release from a Gaussian plume with ground reflection,

$$C = \frac{10^3\,Q}{2\pi\,u\,\sigma_y\sigma_z}\exp\!\left(-\frac{y^2}{2\sigma_y^2}\right)2\exp\!\left(-\frac{H^2}{2\sigma_z^2}\right)\ \mathrm{mg\,m^{-3}}, \tag{3}$$

with Briggs rural coefficients for neutral stability, $\sigma_y = 0.08x(1 + 10^{-4}x)^{-1/2}$ and $\sigma_z = 0.06x(1 + 1.5 \times 10^{-3}x)^{-1/2}$. The borehole [26, 29, 30], Ishigami [26], OTL circuit and piston [31] functions are standard test functions for computer experiments. The flood model [32] gives the maximal annual underflow of a river with truncated Gumbel, truncated normal, triangular and uniform inputs. The borehole model is also run with the input model of Harper and Gupta [29] (normal borehole radius, lognormal radius of influence) and the plume model with GUM-type inputs: lognormal emission rate (median $500\,\mathrm{g\,s^{-1}}$, geometric SD 2), a Weibull wind speed (shape 2, scale $4\,\mathrm{m\,s^{-1}}$, truncated to $[0.5, 15]\,\mathrm{m\,s^{-1}}$) and a normal crosswind offset (SD 15 m). All formulas and input models were checked against the independent implementation in UQTestFuns [27] (relative differences below 10^{-10} for outputs and 10^{-6} for marginal quantiles).

Exceedance is assessed at a fixed threshold where the model has one (the IDLH level for ammonia, $210\,\mathrm{mg\,m^{-3}}$, about 300 ppm, for the plume and CFD models), and otherwise at the reference 95th percentile, for which $P(Y > T) = 0.05$ by construction. Flooding has a probability below 10^{-5} for the flood input model, so its threshold is the reference 5th percentile of the underflow, $P(Y < T) = 0.05$.

3.1 CFD dispersion case

The CFD case replaces the plume formula by a Reynolds-averaged Navier–Stokes (RANS) simulation of a continuous ground-level release upwind of a building, solved with OpenFOAM v1912 [33]. The domain is two-dimensional (a vertical plane along the wind; the building and the release are infinitely wide), from 80 m upstream to 250 m downstream of the windward face and 150 m high. The building is 10 m deep and H_b high; the release is a unit line source 20 m upstream of the building, 0.5 m above the ground. The inflow is the neutral atmospheric surface layer of Richards and Hoxey [34], $U(z) = (u_*/\kappa)\ln((z+z_0)/z_0)$, $k = u_*^2/\sqrt{C_\mu}$, $\varepsilon = u_*^3/(\kappa(z+z_0))$, scaled to the wind speed U_ref at 10 m, with the standard k–ε model ($\sigma_\varepsilon = 1.11$ for consistency with the inflow), a rough-wall function on the ground [35], smooth building walls and a slip top boundary. The concentration s of the passive scalar solves $\nabla\cdot(\mathbf{U}s) - \nabla\cdot((\nu/Sc + \nu_t/Sc_t)\nabla s) = S$, with $Sc = 0.7$ and the turbulent Schmidt number Sc_t as an input, since its appropriate value is uncertain and varies between flows [36]. Flow and scalar are solved together by the SIMPLE algorithm on 7776 hexahedral cells, refined towards the building, until all scaled residuals fall below 10^{-5} (pressure) and 10^{-6} (other fields), or 4000 iterations, typically in 1000–1500 iterations (about 20 s on one core). The output is the concentration at 1.5 m height a distance x_r downstream of the leeward face, $C = 10^3\, Q\, s(x_r)\,\mathrm{mg\,m^{-3}}$ for a release rate Q per metre of crosswind width; since the scalar is passive, x_r and Q need no separate run. The six inputs are uniform: $U_\mathrm{ref} \in [1,6]\,\mathrm{m\,s^{-1}}$, $z_0 \in [0.01, 0.3]$ m, $H_\mathrm{b} \in [6,15]$ m, $Sc_t \in [0.5, 1.3]$, $x_r \in [5,120]$ m and $Q \in [1,20]\,\mathrm{g\,s^{-1}\,m^{-1}}$.

The case contains what the plume formula lacks: flow separation at the building edges, a recirculating wake whose length and strength change with H_b and z_0, receptors inside and beyond the wake, and a discretisation and iteration error that varies with the inputs. It is a benchmark, not a validated prediction of a particular site: the model is *defined* as this discretised RANS solution, which is what a surrogate of an expensive simulation has to reproduce. A reference Monte Carlo of 2×10^5 CFD runs is not affordable. The reference distribution is built instead from 800 runs at independent uniform draws of the four flow inputs, each combined with 50 independent draws of (x_r, Q): 40 000 reference points, of which 4000 (five per run) form the test set and 1600 (two per run) the input set for derived quantities. The grid sensitivity of the case is reported in Section 5.8.

4 Surrogates, designs and protocol

4.1 Surrogates

Gaussian processes. All GPs are fitted on standardised inputs and outputs by maximising the log marginal likelihood (L-BFGS-B, five random restarts) with scikit-learn. The specifications, in order of increasing completeness, are: (i) the *original* kernel of the earlier analysis, a squared exponential with one length scale and an amplitude *fixed* at 1 on the standardised output; (ii) with the amplitude estimated; (iii) with one length scale per input (automatic relevance determination, ARD) but the amplitude fixed; (iv) with both; (v) a Matérn-5/2 ARD kernel with amplitude and a nugget; (vi) (v) plus an amplitude-scaled quadratic polynomial kernel, i.e. a GP with a quadratic trend whose coefficients are integrated out. Length scales are bounded to $[10^{-2}, 10^3]$, the amplitude to $[10^{-3}, 10^3]$ and the nugget variance to $[10^{-8}, 10^{-1}]$. Variants (v) and (vi) are also fitted on the log scale for positive outputs, with logarithmic transformation of inputs whose support is positive.

Sparse polynomial chaos. A sparse PCE [17] with an orthonormal Legendre basis on inputs mapped to $[-1,1]$ (affinely for uniform inputs, through their distribution functions otherwise), candidate sets of

total degree $p = 1, \ldots, 5$ with hyperbolic truncation ($q = 1$ for $p \leq 2$, $q = 0.75$ above), least-angle regression [16] to order the basis terms, ordinary least squares on every active set along the path, and selection of the active set and degree by the corrected LOO error [17]. It is fitted on the raw and on the log scale.

Committees. Nine members: two GPs (squared exponential and Matérn-3/2, both ARD with amplitude), PCEs [14, 15] of degrees 2 and 3 (Bayesian ridge regression), random forests of 50 and 100 trees, a support vector regressor and two neural networks. Members are combined uniformly or with weights $w_k \propto [\mathrm{MSE}_{\mathrm{CV},k}(1+\pi_k)]^{-1}$, with $\mathrm{MSE}_{\mathrm{CV},k}$ the three-fold cross-validation error of member k and $\pi_k \geq 0$ a penalty for predictions below a known physical bound [18, 19].

4.2 Designs

Passive designs are N-point Latin hypercubes drawn on $[0, 1]^d$ and mapped to the inputs by the inverse distribution functions. Adaptive designs start from N_0 Latin hypercube runs ($N_0 = 5$ for $N \leq 20$, $N/4$ above) and add one run at a time, refitting after every run: *GP-SD* adds the candidate with the largest posterior SD [21]; the adaptive committees add the candidate where the active members disagree most, by the variance of their predictions or by the spectral measure UEigV [22] of the earlier analysis; and *AK-MCS* adds the candidate minimising $|\hat{z}(\mathbf{x}) - \varphi(T)|/s(\mathbf{x})$, the learning function of Echard et al. [23], which concentrates runs near the threshold T of the exceedance probability. Candidates are random draws from the input distribution (300 for GP-SD, 2000 for AK-MCS).

4.3 Protocol and statistics

Every strategy is run with the 30 seeds 1000–1029 and budgets $N = 10, 20, 40, 80$ (the full set of strategies at $N = 20$). Seeds are shared and the Latin hypercube is seeded from the global random state, so that for a given seed all passive strategies use the identical design and all adaptive strategies the identical start. Accuracy over the input domain is the root mean square error on 5000 test points divided by the reference output SD (normalised RMSE, nRMSE; 1 corresponds to a constant predictor). The surrogates propagate 3×10^4 input draws; the reference uses 2×10^5 draws of the model (Section 3.1 for the CFD case).

For each strategy and model the median over seeds is reported with a 95% percentile-bootstrap confidence interval (10 000 resamples of the seeds). Two strategies are compared on the same seeds by the median of the per-seed ratio of their errors, with a paired bootstrap 95% confidence interval, and by a two-sided Wilcoxon signed-rank test on the log ratios; the p-values of all comparisons reported are adjusted together by Holm's method. Coverage of prediction and coverage intervals is the share of test points or of the reference distribution inside the interval, averaged over seeds; coverage of intervals for reported quantities is the share of seeds whose interval contains the true value, with a Wilson 95% interval. The interval score [37] combines width and coverage.

5 Results

Table 2 gives the median nRMSE of every strategy on every model at $N = 20$, Table 3 the paired comparisons and Fig. 2 the same relative to the Matérn ARD GP with confidence intervals.

Table 2: Median nRMSE over 30 seeds with $N = 20$ runs; bold: best per model. AK-MCS on the log scale for positive outputs. Generated from the result records; 95% bootstrap intervals are in the supplementary summary.

	GP					Sparse PCE		Committee / adaptive			Targeted
Model	orig.	Matérn	+trend	log	log+tr.	raw	log	CV	GP-SD	adapt.	AK-MCS
Physics	0.591	0.079	0.00018	–	–	**0.00014**	–	0.00018	0.070	0.00015	0.084
Plume	0.602	0.597	0.599	0.418	**0.349**	0.725	0.434	0.582	0.681	0.716	0.385
Borehole	0.295	0.089	0.089	**0.011**	0.014	0.364	0.041	0.217	0.082	0.136	0.013
Ishigami	0.956	0.902	**0.884**	–	–	0.906	–	0.889	0.970	0.895	0.999
OTL circuit	0.198	0.070	0.080	**0.037**	0.045	0.118	0.122	0.121	0.064	0.086	0.051
Piston	0.360	0.264	0.287	0.189	0.174	0.439	0.379	0.303	0.246	0.259	**0.106**
Borehole (GUM inputs)	0.275	0.116	0.117	**0.030**	0.032	0.334	0.309	0.245	0.096	–	0.039
Plume (GUM inputs)	0.696	0.655	0.664	0.365	**0.279**	0.836	0.507	0.664	0.797	–	0.297
Flood	0.227	**0.059**	0.060	–	–	0.210	–	0.109	0.061	–	0.070
CFD dispersion	0.557	0.529	0.500	0.080	**0.060**	0.721	0.208	0.520	0.532	0.448	0.087

Table 3: Paired comparisons at $N = 20$: median over 30 seeds of the ratio of the first strategy's error to the second's (nRMSE; for the exceedance rows, the absolute error of $P(Y > T)$). Below 1 favours the first. Bold: Holm-adjusted $p < 0.05$ (two-sided Wilcoxon signed-rank test on log ratios, all comparisons of the study adjusted together). Generated from the result records.

Comparison	Phys.	Plume	Bore.	Ishi.	OTL	Pist.	Bore.G	Plume.G	Flood	CFD
kernel: amplitude + ARD vs original	**0.14**	0.97	**0.31**	0.95	**0.34**	**0.71**	**0.40**	0.93	**0.25**	0.96
log variables vs raw	–	**0.69**	**0.11**	–	**0.49**	**0.75**	**0.26**	**0.54**	–	**0.17**
quadratic trend vs none	**0.0024**	1.000	1.003	1.000	1.000	1.000	**1.004**	1.000	1.004	1.000
quadratic trend vs none (log)	–	0.89	1.25	–	1.004	1.001	1.04	**0.82**	–	**0.76**
sparse PCE vs GP	**0.002**	**1.23**	**3.83**	1.02	**1.78**	**1.59**	**3.35**	**1.25**	**3.64**	**1.39**
sparse PCE vs GP (log)	–	1.10	**3.48**	–	**3.17**	**2.07**	**11.66**	**1.43**	–	**2.35**
committee vs GP + trend	0.995	0.96	**2.36**	1.02	**1.67**	1.16	**2.21**	1.002	**1.90**	1.08
CV weights vs uniform	**0.00047**	0.98	**0.84**	0.991	**0.54**	**0.86**	0.82	0.994	**0.38**	0.96
adaptive vs passive GP	1.007	1.18	0.94	1.06	0.86	1.01	0.96	1.22	1.06	0.93
adaptive vs passive committee	**0.80**	**1.25**	**0.61**	0.98	**0.70**	0.84	–	–	–	**0.84**
UEigV vs variance	1.02	0.90	**1.67**	1.07	**1.72**	**1.26**	–	–	–	–
AK-MCS vs passive, exceedance	**0.47**	0.88	**0.11**	0.88	**0.29**	**0.31**	**0.22**	1.41	0.35	–
AK-MCS vs passive, exceedance (log)	–	1.13	0.78	–	0.65	**0.17**	**0.27**	**0.35**	–	0.94
log vs raw, exceedance	–	0.39	**0.21**	–	**0.59**	0.85	**0.19**	0.43	–	0.50

5.1 GP specification and variable transformation

Estimating the kernel amplitude and one length scale per input (Matérn ARD versus the original kernel) reduced the nRMSE on six of the nine models, by factors of 0.14 (physics) to 0.71 (piston), and did not change it on the plume (both input models) and Ishigami models, where neither kernel can resolve the response with 20 runs. On the physics model the original kernel, with its amplitude fixed at the variance of the standardised output, cannot represent a quadratic bowl whose values far exceed one standard deviation and reverts towards the mean between runs. Estimating the amplitude alone reduced the median nRMSE from 0.59 to 0.033, eighteen-fold; the earlier analysis had attributed this error to the size of the input domain (Section 6.1).

Fitting on the log scale, with log-transformed positive inputs, reduced the nRMSE on all six models with positive outputs, by factors of 0.11 (borehole) to 0.75 (piston); all six reductions remain significant after the Holm adjustment. The GP then interpolates a nearly additive function of the log inputs, which is what products and power laws such as the plume, borehole and OTL formulas become.

Adding a quadratic trend to the GP reduced the nRMSE on the physics model by a factor of 400, to the level of the sparse PCE, and changed it by less than 0.5% on every other model: a GP with a trend loses practically nothing when the trend is absent. On the log scale, the trend improved the plume models (0.42 to 0.35 with uniform inputs, 0.37 to 0.28 with GUM-type inputs), whose log concentration is close to quadratic in the inputs.

5.2 Sparse PCE and committees

Sparse PCE matched the trend GP on the physics model and the Matérn GP on the Ishigami model, and was 1.2 to 3.8 times worse than the GP on the other seven models (1.1 to 12 times on the log scale); with 20 runs and up to eight inputs, the corrected LOO error selects low-degree expansions that cannot follow the borehole, OTL and flood responses as closely as the GP.

The CV-weighted committee was never better than the GP with a quadratic trend on the same design. It tied on five models and was 1.7 to 2.4 times worse on the other four: the borehole (both input models), OTL and flood models. Its advantage on the physics model in the earlier analysis is entirely that of its degree-2 PCE member, which the trend GP reproduces (ratio 1.00). CV weighting remained necessary for a committee: it was never worse than uniform averaging and better on five models, by up to three orders of magnitude on the physics model, where uniform averaging dilutes the exact member with poor ones.

5.3 Adaptive and targeted designs

Space-filling adaptive designs gave no consistent gain at $N = 20$. Adding runs at the largest GP posterior SD never changed the nRMSE significantly (ratios 0.86–1.22). The adaptive committee helped on the physics, borehole and OTL models (0.61–0.80) and hurt on the plume model (1.25), where its estimate of $P(C > \text{IDLH})$ was off by a median of 0.17, against 0.02 for the passive committee. The spectral criterion UEigV was never better than plain committee variance and was worse on three models (1.26–1.72).

Targeting the threshold changes the picture for exceedance probabilities. AK-MCS reduced the median absolute error of $P(Y > T)$ significantly in 8 of the 15 comparisons, by factors of 0.11 to 0.47, and was never significantly worse. The largest gains were on the piston model (0.0099 to 0.0014 on the log scale), the borehole model (0.0091 to 0.0009 on the raw scale) and the GUM-type input models. Targeting comes at a cost elsewhere: on the Ishigami model the AK-MCS design raised the median error of the propagated mean from 8.5% to 28%, because its runs cluster near the threshold.

5.4 Propagated quantities

Table 4 gives the validation errors of the reported quantities for the GP recommended for each model and for the committee. With the recommended GP, the 95th percentile was within 0.2% (OTL, flood) to 16% (Ishigami) of the reference, and $P(Y > T)$ within 0.0013 to 0.036 of its reference value. The committee's 95th percentile was off by 6.9–31% on six of the nine models. At the IDLH threshold of the plume model, the log-variable GP estimated $P(C > 210\,\text{mg}\,\text{m}^{-3}) = 0.123$ with a median absolute error of 0.009, against 0.023 for the raw-variable GP.

5.5 Selecting the surrogate from the runs alone

The procedure of Fig. 6 fits the candidates (Matérn GP, GP with quadratic trend and sparse PCE, and their log-scale versions for positive outputs) and keeps the one with the smallest LOO RMSE, computed on the original scale of Y from the N runs alone. Applied to every seed, this choice had a median nRMSE between 1.000 and 1.044 times that of the best candidate in hindsight on the nine analytical models, and was the best candidate itself in 27–77% of seeds; the largest shortfalls were on the Ishigami and plume models, where several candidates are almost equally poor or equally good. Relative to always using the Matérn GP, the selection reduced the nRMSE by a factor of 500 on the physics model (where it chose the sparse PCE or the trend GP), by factors of 0.12–0.84 on the models with positive outputs (choosing a

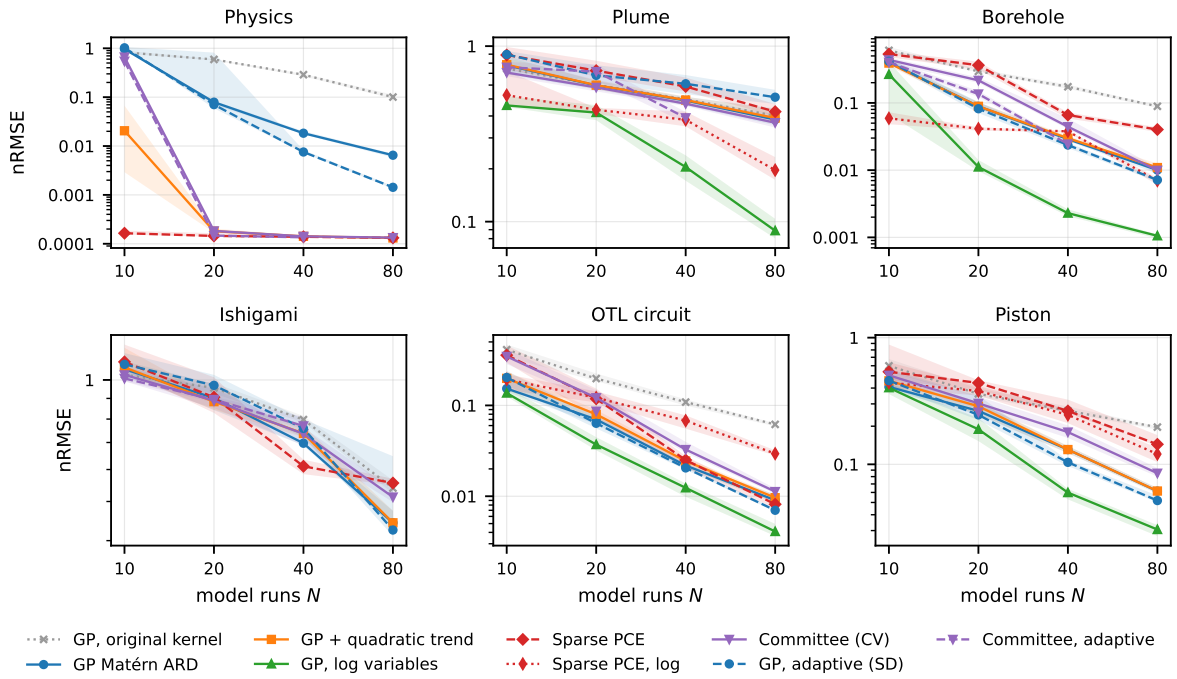


Figure 1: Median nRMSE over 30 seeds against the number of runs N for the six models with uniform inputs; shaded: 95% bootstrap confidence interval of the median. Adaptive committees were run up to $N = 40$.

log-scale GP in most seeds), and left it unchanged on the flood and Ishigami models. The selection step therefore recovers practically all of the benefit of the right specification without access to the true model.

5.6 Dependence on the number of runs

Fig. 1 repeats the main strategies with $N = 10$, 40 and 80 runs on the six models with uniform inputs. The ranking of the specifications is stable: the original kernel stays the worst GP at every budget, and the benefit of the log scale grows with N, from factors of 0.62–0.92 at $N = 10$ to 0.10–0.48 at $N = 80$ on the four positive models, because the transformed response is smooth enough to be learned to high accuracy once the runs resolve it. Sparse PCE remains best on the physics model at every budget (at $N = 10$ it already has an nRMSE of 1.6×10^{-4}, against 0.97 for the Matérn GP) and worse than the GP on most of the others.

The verdict on space-filling adaptive designs depends on the budget. At $N = 10$ and 20, adding runs at the largest GP posterior SD never helped significantly. At $N = 40$ and 80 it reduced the nRMSE on the physics (0.22–0.43), borehole (0.77), OTL (0.75 at $N = 80$) and piston (0.81–0.87) models, which are smooth and on which the GP's uncertainty is informative once its hyperparameters are well determined, but increased it on the plume model (1.19 at $N = 40$), whose error concentrates in the high-concentration corner that the posterior SD does not single out. The adaptive committee was better than the passive one at $N = 40$ on the physics and borehole models (0.94 and 0.50), no different elsewhere, and no better at $N = 10$. Adaptive selection is thus worth considering when the budget is several times the number of inputs, but not as a remedy for very small budgets, when the surrogate's own uncertainty is itself poorly estimated.

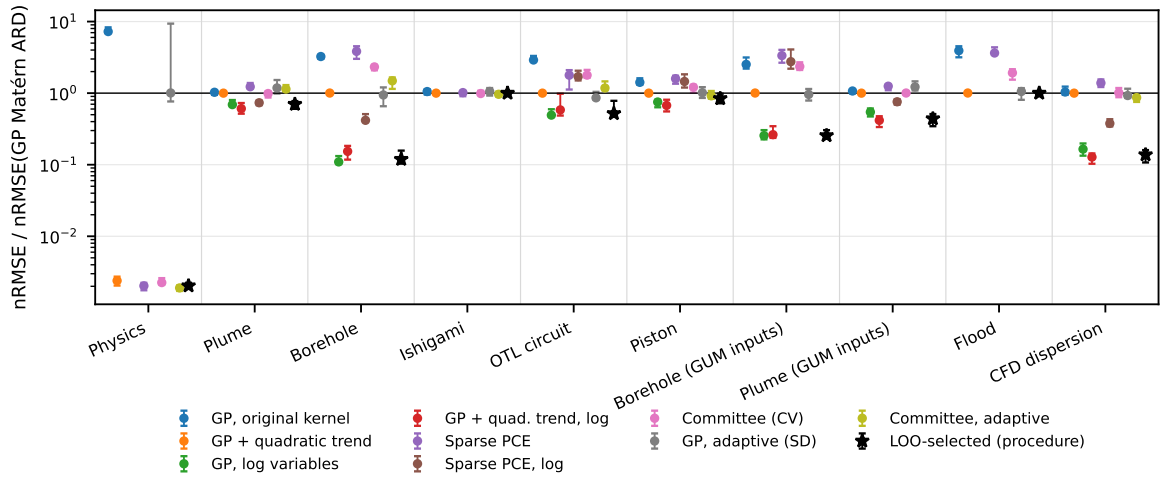


Figure 2: nRMSE of each strategy relative to the Matérn ARD GP on the same design, $N = 20$: median over 30 seeds of the paired ratio with 95% bootstrap interval. Star: the candidate chosen per seed by the smallest LOO error (the procedure of Fig. 6).

Table 4: Validation error of the reported quantities, median over 30 seeds, $N = 20$: relative error (%) of the mean, SD and $y_{0.95}$ and absolute error ($\times 100$) of $P(Y > T)$ for the recommended GP (log scale for positive outputs), the same with an AK-MCS design, and the CV-weighted committee. Generated from the result records.

	GP (passive)				AK-MCS	Committee	
Model	mean	SD	$y_{0.95}$	P	P	$y_{0.95}$	P
Physics	1.36	4.22	3.88	1.37	0.77	0.41	0.15
Plume	4.74	15.9	6.94	0.91	1.13	23.1	2.36
Borehole	0.17	0.58	0.60	0.18	0.13	10.9	3.05
Ishigami	8.48	18.8	16.0	3.61	3.19	28.7	2.89
OTL circuit	0.17	0.60	0.21	0.14	0.14	2.29	1.31
Piston	0.62	2.75	1.91	0.99	0.14	7.47	2.68
Borehole (GUM inputs)	0.09	1.22	0.47	0.13	0.04	6.85	2.25
Plume (GUM inputs)	6.27	17.0	8.76	0.80	0.29	31.2	3.72
Flood	0.06	1.81	0.18	0.41	0.17	0.75	1.16
CFD dispersion	4.41	7.62	6.89	0.85	0.90	27.1	5.86

5.7 Is the surrogate contribution valid?

Fig. 3 and Table 5 test the three models of the surrogate error of Section 2.2 against the true models, for every surrogate fitted to 20 runs (48 combinations of model and surrogate on the nine analytical models, 30 seeds each).

Pointwise intervals. The jackknife+ interval covered between 95.2% and 99.0% of the test points (median 97.4%) in every combination, in line with its distribution-free guarantee of at least 90%. The CV-normal interval covered 89.7–97.0% (median 94.2%). The GP's own predictive interval covered 68–98% (median 89%) and fell below 80% on the plume (both input models), Ishigami and piston models, where the GP is least accurate: its posterior SD reflects the fitted covariance, not the actual error. Validity has a price in width: jackknife+ intervals were a median 1.6 (1.2–2.9) times as wide as CV-normal ones, and the narrow GP intervals still had the best interval score [37] in 22 of the 33 GP combinations. For a measurement uncertainty statement, which claims a coverage probability, the coverage is the property that matters.

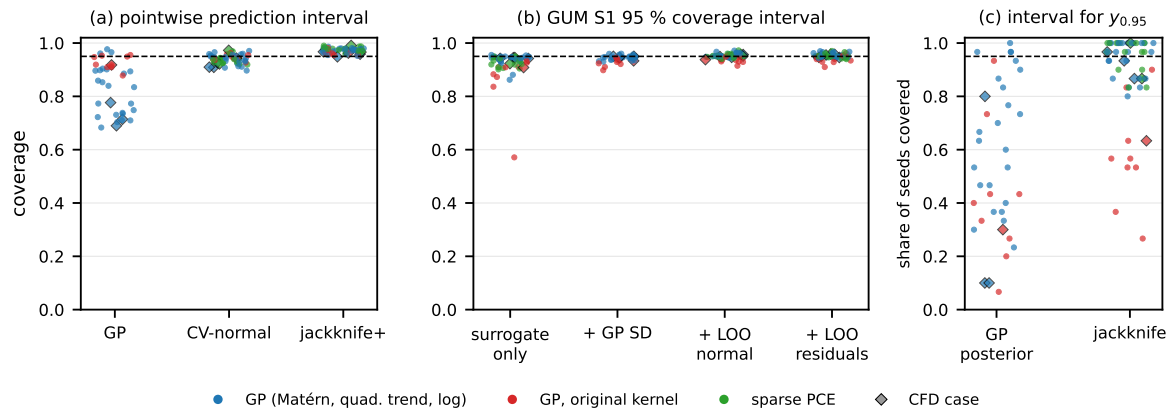


Figure 3: Coverage of nominal 95% intervals, mean over 30 seeds; one marker per combination of model and surrogate (circles: the nine analytical models; diamonds: the CFD case, Section 5.8). (a) Pointwise prediction intervals on 5000 test points. (b) Achieved coverage of the GUM S1 95% coverage interval of Y under the reference distribution, with the surrogate treated as exact or with its error added to each trial. (c) Share of seeds whose 95% interval for $y_{0.95}$ contains the true value. Dashed line: 0.95.

Table 5: Coverage (%) of nominal 95% intervals for the GP recommended for each model (log variables for positive outputs), 30 seeds, $N = 20$. Pointwise: share of test points. GUM S1: achieved coverage of the 95% coverage interval of Y. $y_{0.95}$, $P(Y > T)$: share of seeds whose interval contains the true value (GP: joint posterior paths; J: jackknife). Generated from the result records.

	Pointwise			GUM S1 interval of Y				$y_{0.95}$		$P(Y > T)$	
Model	GP	CV	J+	exact	+LOO N	+LOO res.	+GP	GP	J	GP	J
Physics	98	96	97	92	95	95	94	90	87	87	70
Plume (log)	71	91	97	94	95	95	94	77	93	70	97
Borehole (log)	97	96	97	95	95	95	95	97	100	97	97
Ishigami	71	92	97	86	97	97	93	73	83	73	73
OTL circuit (log)	90	95	98	95	95	95	95	93	100	93	100
Piston (log)	77	91	95	94	94	94	94	63	93	60	90
Borehole (GUM inputs) (log)	96	96	97	95	95	95	95	97	100	97	100
Plume (GUM inputs) (log)	73	93	98	94	95	95	94	60	100	57	100
Flood	90	95	98	95	95	95	95	97	100	73	100
CFD dispersion (log)	78	91	95	95	95	95	95	80	97	87	97

GUM S1 coverage interval. Treating the surrogate as exact gave coverage intervals for Y whose achieved coverage under the reference distribution ranged from 57% to 96% (median 94%), and standard uncertainties $u(y)$ that were a median 0.92 times the reference (down to 0.41). Adding the LOO error to every Monte Carlo trial, equation (1), raised the achieved coverage to 91.5–97.3% (median 95.0%) as a normal error and to 91.0–97.0% (median 94.9%) by resampling the residuals, with $u(y)$ a median 1.00 (0.78–1.29) times the reference. The GP's predictive SD gave 89.8–96.0% (median 94.5%). The largest correction occurred where it is needed most: for the GP with the original kernel on the Ishigami model, coverage rose from 57% to 95%. Adding the error did not systematically improve exceedance probabilities, which depend on the tail of the surrogate rather than on its overall spread.

Reported quantities. The difference between the approaches is largest for the quantities a GUM S1 evaluation reports. Intervals for $y_{0.95}$ from joint GP posterior sample paths contained the true value in a median 53% of seeds (7–100%), and those for $P(Y > T)$ in 57%: the GP posterior describes the uncertainty of the interpolant as the fitted covariance model sees it, and is overconfident wherever that model is misspecified. The jackknife intervals contained the true value in a median 97% of seeds for both quantities, with the lowest values for the original-kernel GP, whose error is a bias that no resampling of its own fits can reveal.

5.8 CFD dispersion case

The case. Fig. 4 shows the flow and concentration field for the centre of the input box: the plume is lifted over the building, part of it is trapped in a recirculating wake that extends several building heights downstream, and the concentration at breathing height is nearly uniform inside the wake and decays beyond it. Of the 2900 CFD runs of the study (1550 for the designs and the reference, the rest added by the adaptive designs), 80% met the residual criteria within 4000 iterations (median 1322 iterations); for the others, mostly tall buildings whose steady wake keeps the residuals oscillating, the receptor concentration varied by less than 0.16% over the last 2000 iterations, so the iteration error is negligible. Refining the mesh twice in each direction changed the receptor concentration by 0.9–6.5% (median 3.7%) in six flow states spread over the input box: a discretisation error that is part of the model as defined, as it would be for any production CFD model. The reference gives a mean of $434\,\mathrm{mg\,m^{-3}}$, an SD of $413\,\mathrm{mg\,m^{-3}}$, a 95th percentile of $1252\,\mathrm{mg\,m^{-3}}$ and $P(C > \mathrm{IDLH}) = 0.673$, with standard uncertainties from a bootstrap over the 800 reference runs of 2.3%, 3.4%, 3.3% and 0.007; validation errors of the propagated quantities below about 5% are therefore not resolved.

Specification and selection. The CFD case confirms the analytical results (Table 2, last row). Kernel specification did not matter (ratio 0.96), but the log scale did: the log-variable GP had a median nRMSE of 0.080 against 0.53 for the Matérn GP (ratio 0.17, better in all 30 seeds), and a quadratic trend on the log scale reduced it further to 0.060 (ratio 0.76). Sparse PCE was 1.4 times worse than the GP on the raw scale and 2.3 times on the log scale, and the CV-weighted committee was no better than the GP with trend (ratio 1.08). The selection by LOO error chose a log-scale GP in all 30 seeds and was within 0.1% of the best candidate in hindsight. With the log-variable GP, the 95th percentile was within 6.9% of the reference and $P(C > \mathrm{IDLH})$ within 0.009, against 18.5% and 0.019 for the Matérn GP and 27% and 0.059 for the committee.

Adaptive designs. Adding runs at the largest GP posterior SD did not change the nRMSE (ratio 0.93), and AK-MCS did not improve $P(C > \mathrm{IDLH})$ (ratio 0.94): the IDLH is exceeded in two thirds of the cases, so the threshold lies in the bulk of the distribution rather than in a tail, where a targeted design has most to gain. The adaptive committee reduced the nRMSE relative to the passive one (ratio 0.84) but, as on the plume model, its estimate of $P(C > \mathrm{IDLH})$ was off by a median of 0.11, against 0.06 for the passive committee and 0.009 for the log-variable GP.

Surrogate contribution. The coverage results carry over to the CFD case (Table 5, last row; diamonds in Fig. 3). Pointwise, the GP intervals covered 69–92% of test points and the jackknife+ intervals 94.8–99.1%. Treating the surrogate as exact gave GUM S1 coverage intervals with an achieved coverage of 90.8–94.6%; adding the LOO error gave 93.8–95.7%. The GP posterior was most misleading for the 95th percentile: with the raw-scale GPs (Matérn, with or without trend) its 95% interval contained the true value in only 10% of seeds, against 87–93% for the jackknife; with the log-variable GP the figures were 80% and 97%.

6 Discussion

6.1 Relation to the earlier analysis

The earlier analysis made five claims. Three are withdrawn or revised. The catastrophic GP failure on large domains is a kernel-specification artefact: with the amplitude estimated, the GP no longer reverts to the mean (Fig. 5). The spectral acquisition criterion gives no advantage over plain committee variance on any of the six models on which it was tested, and is worse on three. The committee's advantage is not general but comes from a member that matches the dominant term, which a GP with a quadratic trend

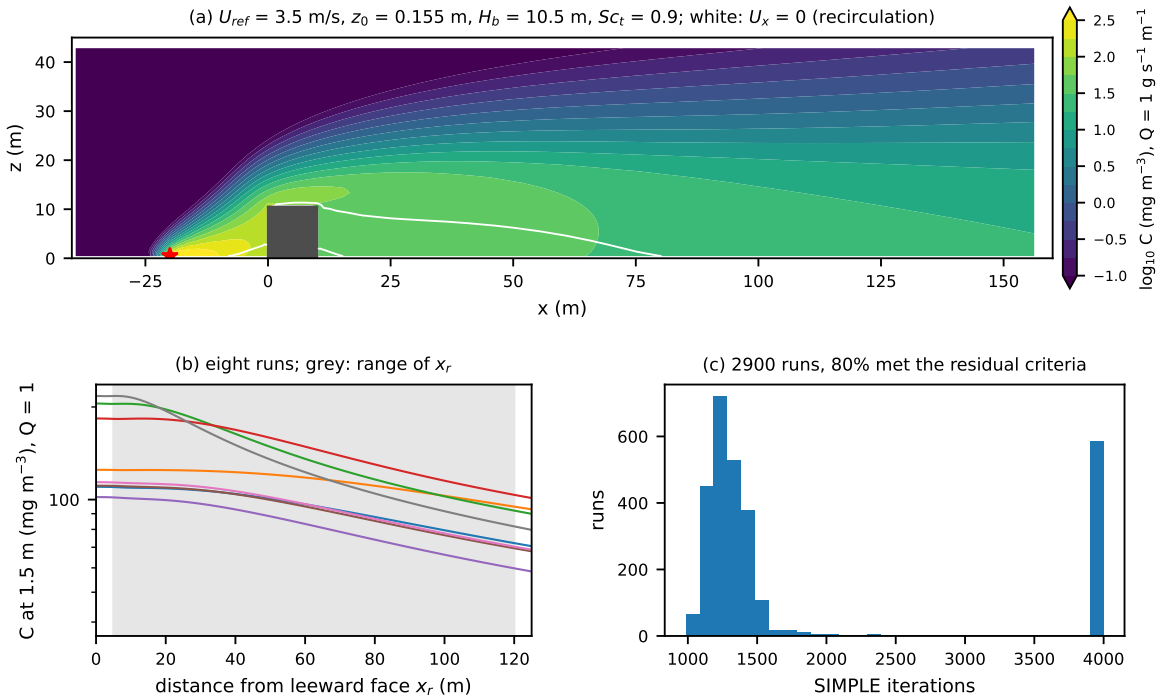


Figure 4: CFD dispersion case. (a) Concentration at a unit release rate for the centre of the input box; white line: zero streamwise velocity, enclosing the recirculating wake; star: release. (b) Concentration at 1.5 m height against distance from the leeward face for eight reference runs. (c) Iterations of the 2900 runs; the runs at 4000 did not meet the residual criteria but changed by less than 0.16% over their last 2000 iterations.

or a sparse PCE reproduces on its own. Two claims survive: aggregation must be error-weighted, and an adaptive committee can be more accurate than a passive one: it was on four of the seven models on which it was run, including the CFD case, although it remained less accurate than the log-variable GP wherever the output is positive, and its exceedance probabilities were worse on the plume and CFD cases. The records of the earlier campaigns are kept in the repository [28].

6.2 What the surrogate contribution should be

The coverage results separate two uses of a surrogate's uncertainty. Pointwise, the GP posterior is a useful but uncalibrated description: it locates where the surrogate is uncertain, which is why it drives good adaptive designs at larger budgets, but its scale is set by a covariance model fitted to 20 runs and is too small wherever that model is wrong. For an uncertainty statement the scale is what matters, and the leave-one-out residuals estimate it directly, without trusting the covariance model; the jackknife+ turns them into intervals with a guarantee. The difference is largest for the reported quantities. A GUM S1 coverage interval of Y is fairly robust, because surrogate error that varies across the input space partly averages out in the distribution of Y, and adding the LOO error to each trial restores its nominal coverage. A 95th percentile or an exceedance probability is not robust: it depends on the surrogate in a small region. Intervals from the GP posterior contained the true value in only about half of the seeds (median over model and surrogate), whereas the jackknife over LOO fits did so in a median 97% of seeds. We therefore recommend the LOO error as the surrogate contribution, and the jackknife over LOO fits for the uncertainty of quantities derived from the propagated distribution.

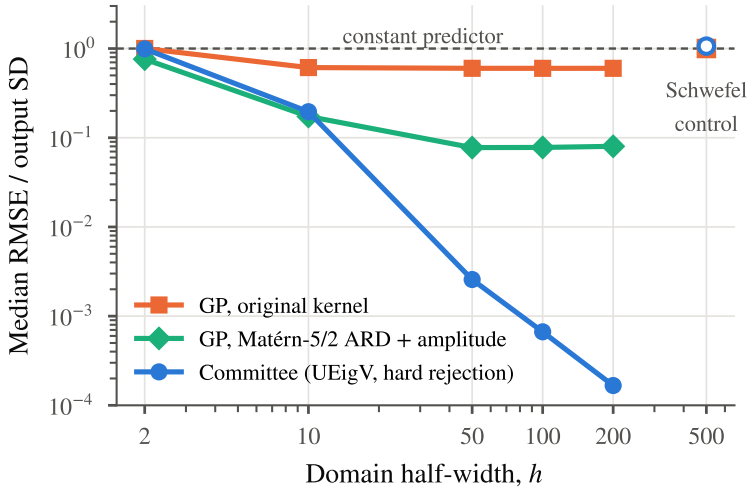


Figure 5: Physics model on $[-h, h]^3$, $N = 20$: median nRMSE over 30 seeds against the domain half-width, for the GP with the original kernel (amplitude fixed), the Matérn ARD GP with amplitude, and the committee of the earlier analysis. Open markers: Schwefel function on $[-500, 500]^3$. Dashed line: constant predictor.

6.3 Limitations

The study covers ten models with up to eight inputs and budgets of 10 to 80 runs; higher-dimensional problems, where sparse and additive structures matter more, may rank the surrogates differently. Two models were run with a single budget per input distribution, and the adaptive committees were run only up to $N = 40$ because of their cost. The CFD case is two-dimensional and steady, with a standard k–ε model; it is a benchmark with the features a dispersion simulation has, not a validated prediction for a site, and its reference distribution is built from 800 runs rather than 2×10^5. The coverage results are averages over seeds: an individual evaluation can still be unlucky, and the jackknife+ guarantee (at least 90% for nominal 95%) holds on average over designs, not for every design. The log transformation applies only to positive outputs, and its back-transformation to the predictive median slightly underestimates the mean of a skewed output; the bias is included in the validation errors reported. Finally, the paired tests are exploratory: all p-values were Holm-adjusted together, and nominal differences that do not survive the adjustment are not claimed.

7 A procedure for surrogate-based GUM S1 evaluations

Fig. 6 condenses the results into a procedure that needs only the N runs. Its two key steps were validated on the ten models: the selection of the surrogate by its LOO error recovered practically all the benefit of the best specification (Section 5.5), and the LOO error as surrogate contribution restored the nominal coverage of the GUM S1 coverage interval (Section 5.7). Candidates should include a GP with an estimated amplitude, one length scale per input and a quadratic trend, and a sparse PCE, each on the log scale when the output is positive. Adaptive designs are optional: a targeted design such as AK-MCS is worthwhile when a threshold decision is the purpose and the threshold lies in a tail of the distribution, and variance-driven designs when the budget is several times the number of inputs and the model is smooth.

8 Conclusion

For GUM S1 evaluations in which the measurement model can be run only a few tens of times, the way a surrogate is specified matters more than the strategy around it. A GP should estimate its amplitude

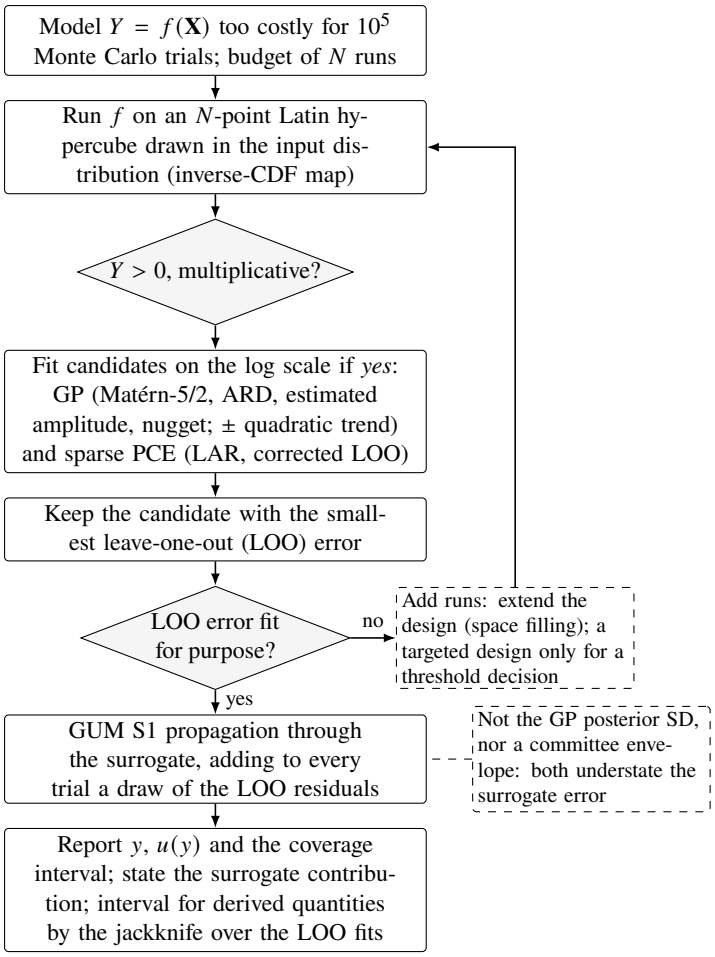


Figure 6: Procedure for a GUM S1 evaluation with a surrogate fitted to N runs of an expensive model.

and one length scale per input, include a quadratic trend, and work on the log scale when the output is positive; a failure of GP surrogates reported in an earlier analysis was an artefact of one hyperparameter held fixed. Among such candidates and a sparse PCE, the one with the smallest leave-one-out error was as good as the best in hindsight on all ten models, including a CFD simulation of a release behind a building. Committees of surrogates and variance-driven adaptive designs gave no consistent gain at 20 runs, whereas a design targeted at the threshold improved exceedance probabilities when the threshold lay in a tail of the distribution. Finally, the surrogate's own error must be part of the uncertainty statement, and it should be estimated from leave-one-out residuals rather than from the GP's predictive variance: the former restored the nominal coverage of the GUM S1 coverage interval and gave valid intervals for reported quantities, the latter was overconfident precisely where the surrogate was poor.

Data and code availability

All code, the per-seed records of every strategy, the cached CFD runs, figure scripts and an analysis script that recomputes every table and figure are available at `https://github.com/Yugam2508/fyp-active-learning-cfd` [28]; `docs/REPRODUCIBILITY.md` lists the commands and the software versions.

CRediT authorship contribution statement

Yugam Jinesh Vora: conceptualisation, methodology, software, investigation, formal analysis, visualisation, writing – original draft. Wee Hoe Ng: conceptualisation, supervision, writing – review and editing.

Declaration of competing interest

The authors declare no competing interests.

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